

Lecture-09 [24 August 2026]

Bernstein's inequality

Fix $n \in \mathbb{N}$. Suppose that $X_1, \dots, X_n$ are independent and sub-exponential.

Fix $a_1, \dots, a_n \in \mathbb{R}$. Without loss of generality, let $\mathbb{E}[X_k] = 0 \quad \forall k$

Then,

$$\mathbb{P}\left(\left|\sum_{k=1}^n a_k X_k\right| \geq \varepsilon\right) \leq 2 \exp\left(-\alpha \min\left\{\frac{\varepsilon^2}{\sum_{k=1}^n a_k^2 \|X_k\|_{\Psi_1}^2}, \frac{\varepsilon}{\max_k |a_k| \|X_k\|_{\Psi_1}}\right\}\right)$$

$\alpha > 0$ is a universal constant

for small ε , decay is $e^{-O(\varepsilon^2)}$
for large ε , decay is $e^{-O(\varepsilon)}$

Proof of Bernstein's inequality

Because X_k is sub-exponential for each k , we have

$$\mathbb{E}\left[e^{t a_k X_k}\right] \leq e^{\frac{c^2}{2} t^2 a_k^2 \|X_k\|_{\Psi_1}^2} \quad \forall t: |t| \leq \frac{1}{c_k |a_k| \|X_k\|_{\Psi_1}}$$

$$\Rightarrow \mathbb{E}\left[e^{t a_k X_k}\right] \leq e^{\frac{c^2}{2} t^2 a_k^2 \|X_k\|_{\Psi_1}^2} \quad \forall t: |t| \leq \frac{1}{c B},$$

where

$$c \geq \max_k c_k \quad \text{and} \quad B = \max_k |a_k| \|X_k\|_{\Psi_1}.$$

$\hookrightarrow$ we can choose c so that $c > 1$

We then have

$$\begin{aligned} \mathbb{E}\left[e^{t \sum_{k=1}^n a_k X_k}\right] &= \prod_{k=1}^n \mathbb{E}\left[e^{t a_k X_k}\right] \\ &\leq e^{\frac{c^2}{2} t^2 \sum_{k=1}^n a_k^2 \|X_k\|_{\Psi_1}^2} \\ &= e^{\frac{c^2}{2} t^2 \underbrace{\sum_{k=1}^n a_k^2 \|X_k\|_{\Psi_1}^2}_{=: V}} \end{aligned} \quad \forall t: |t| \leq \frac{1}{c B}$$

Using the Chernoff bounding technique,

$$\mathbb{P}\left(\sum_{k=1}^n a_k X_k \geq \varepsilon\right) \leq \inf_{\substack{t > 0: \\ |t| \leq \frac{1}{c B}}} \frac{\mathbb{E}\left[e^{t \sum_{k=1}^n a_k X_k}\right]}{e^{t\varepsilon}}$$

$$\leq \inf_{0 < t \leq \frac{1}{cB}} e^{-(t\varepsilon - c^2 t^2 V)}$$

The optimum is achieved at $t^* = \frac{\varepsilon}{2c^2 V}$. However, this may or may not be in the range $(0, \frac{1}{cB}]$.

We set

$$t_\varepsilon := \min \left\{ \frac{\varepsilon}{2c^2 V}, \frac{1}{2cB} \right\}.$$

Clearly, $t_\varepsilon \in (0, \frac{1}{cB}]$. Also,

$$t_\varepsilon \leq \frac{\varepsilon}{2c^2 V} \Rightarrow c^2 t_\varepsilon^2 V \leq \frac{t_\varepsilon \varepsilon}{2}$$

$$\Rightarrow t_\varepsilon \varepsilon - c^2 t_\varepsilon^2 V \geq \frac{t_\varepsilon \varepsilon}{2}$$

Hence,

$$\begin{aligned} \mathbb{P} \left(\sum_{k=1}^n a_k X_k \geq \varepsilon \right) &\leq e^{-(t_\varepsilon \varepsilon - c^2 t_\varepsilon^2 V)} \\ &\leq e^{-t_\varepsilon \varepsilon / 2} \end{aligned}$$

$$= \exp \left(- \min \left\{ \frac{\varepsilon^2}{4c^2 V}, \frac{\varepsilon}{4cB} \right\} \right)$$

$$c > 1 \Rightarrow c \leq c^2 \checkmark \leq \exp \left(- \frac{1}{4c^2} \min \left\{ \frac{\varepsilon^2}{V}, \frac{\varepsilon}{B} \right\} \right) \rightarrow \textcircled{a}$$

Along similar lines, noting that

$$\| -X_k \|_{\Psi_1} = \| X_k \|_{\Psi_1},$$

We get

$$\mathbb{P} \left(\sum_{k=1}^n a_k X_k \leq -\varepsilon \right) \leq \exp \left(- \frac{1}{4c^2} \min \left\{ \frac{\varepsilon^2}{V}, \frac{\varepsilon}{B} \right\} \right) \rightarrow \textcircled{b}$$

Combining $\textcircled{a}$ and $\textcircled{b}$, we get the desired result.
(with $\alpha = \frac{1}{4c^2}$)

→ $\|\underline{x}\|_2^2$ is sub-exponential

Towards the end of the previous lecture, we saw that if

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, then for all $\delta > 0$,

$$\mathbb{P}\left(\left|\|\underline{x}\|_2^2 - n\sigma^2\right| \geq n\delta\right) \leq 2 \exp -c' \min\left\{\frac{n\delta^2}{\sigma^4}, \frac{n\delta}{\sigma^2}\right\}$$

for some universal constant $c' > 0$.

Suppose that $\underline{x} = (X_1, \dots, X_n)$ has independent, sub-Gaussian components

What can we say about the behaviour of $\|\underline{x}\|_2$?

sub-exponential ← ? → sub-Gaussian?

The below result shows that $\|\underline{x}\|_2$ is sub-Gaussian.

Proposition: Let $X_1, \dots, X_n$ be independent and sub-Gaussian. Furthermore, let

$$\mathbb{E}[X_i] = 0, \quad \text{Var}(X_i) = 1, \quad K_i := \|X_i\|_{\Psi_2} \quad \forall i.$$

Let $K := \max_i K_i$. Set $\underline{x} = (X_1, \dots, X_n)$. Then:

① There exists a universal $c > 0$ such that

$$\mathbb{P}\left(\left|\|\underline{x}\|_2 - \sqrt{n}\right| > \varepsilon\right) \leq 2 \exp\left(\frac{-c\varepsilon^2}{K^4}\right) \quad \forall \varepsilon > 0,$$

thus proving that $\|\underline{x}\|_2 - \sqrt{n}$ is sub-Gaussian.

② There exists a universal $c' > 0$ such that

$$\left\|\|\underline{x}\|_2 - \sqrt{n}\right\|_{\Psi_2} \leq c' K^2 = c' \max_i \|X_i\|_{\Psi_2}^2.$$

Proof of proposition

Let $Z_i = X_i^2 - 1$. Then, $Z_1, \dots, Z_n$ are mutually independent and $\mathbb{E}[Z_i] = 0 \quad \forall i$.

We know that if X_i is sub-Gaussian, then X_i^2 is sub-exponential

$$\|1\|_{\Psi_1} = \inf \{k > 0: e^{1/k} \leq 2\} = \frac{1}{\log 2}$$

$\Rightarrow Z_i$ is sub-exponential $\forall i$.

Moreover,

$$\begin{aligned} \|Z_i\|_{\Psi_1} &= \|X_i^2 - 1\| \stackrel{\text{triangle inequality}}{\leq} \|X_i^2\|_{\Psi_1} + \|1\|_{\Psi_1} \\ &= \|X_i^2\|_{\Psi_1} + \frac{1}{\log 2} \\ &= \|X_i\|_{\Psi_2}^2 + \frac{1}{\log 2} \quad \left(\because \|X_i^2\|_{\Psi_1} = \|X_i\|_{\Psi_2}^2 \right) \\ &\leq 2 \|X_i\|_{\Psi_2}^2, \quad \left(\frac{1}{\log 2} \leq \|X_i\|_{\Psi_2}^2 \right) \end{aligned}$$

Where the last line follows by noting that

$$\begin{aligned} 2 &\geq \mathbb{E} \left[e^{X_i^2 / \|X_i\|_{\Psi_2}^2} \right] \\ &\geq e^{\mathbb{E}[X_i^2] / \|X_i\|_{\Psi_2}^2} \quad \left(\text{Jensen's inequality on } f(x) = e^x \right) \\ &= e^{1 / \|X_i\|_{\Psi_2}^2} \Leftrightarrow \|X_i\|_{\Psi_2}^2 \geq \frac{1}{\log 2}. \end{aligned}$$

By Bernstein's inequality, we get

$$\mathbb{P} \left(\left| \sum_{i=1}^n Z_i \right| \geq \varepsilon \right) \leq 2 \exp \left(-c' \min \left\{ \frac{\varepsilon^2}{\sum_{i=1}^n \|Z_i\|_{\Psi_1}^2}, \frac{\varepsilon}{\max_i \|Z_i\|_{\Psi_1}} \right\} \right)$$

Using the relation

$$\|Z_i\|_{\Psi_1} \leq 2 \|X_i\|_{\Psi_2}^2 \leq 2K^2, \text{ we get}$$

$$\begin{aligned} \mathbb{P} \left(\left| \sum_{i=1}^n Z_i \right| \geq \varepsilon \right) &\leq 2 \exp \left(-c' \min \left\{ \frac{\varepsilon^2}{4nK^4}, \frac{\varepsilon}{2K^2} \right\} \right) \\ \Leftrightarrow \mathbb{P} \left(\left| \sum_{i=1}^n X_i^2 - n \right| \geq \varepsilon \right) &\leq 2 \exp \left(-c' \min \left\{ \frac{\varepsilon^2}{4nK^4}, \frac{\varepsilon}{2K^2} \right\} \right) \\ \Leftrightarrow \mathbb{P} \left(\left| \|\underline{X}\|_2^2 - n \right| \geq \varepsilon \right) &\leq 2 \exp \left(-c' \min \left\{ \frac{\varepsilon^2}{4nK^4}, \frac{\varepsilon}{2K^2} \right\} \right). \end{aligned}$$

$\forall \varepsilon > 0$.

Exercise: Show that for any $z \in \mathbb{R}$ and $\delta > 0$,

guided exercise

$$|z-1| > \delta \Rightarrow |z^2-1| > \max\{\delta, \delta^2\}.$$

Using the above exercise, we get

$$\begin{aligned} \mathbb{P}(|\|X\|_2 - \sqrt{n}| > \varepsilon) &\leq \mathbb{P}(\|X\|_2^2 - n \geq n \max\left\{\frac{\varepsilon}{\sqrt{n}}, \frac{\varepsilon^2}{n}\right\}) \\ &\leq 2 \exp\left(-c' \min\left\{\frac{n^2 \max\{\varepsilon^2/n, \varepsilon^4/n^2\}}{4nK^4}, \frac{n \max\{\varepsilon/\sqrt{n}, \varepsilon^2/n\}}{2K^2}\right\}\right) \\ \boxed{\min\left\{\max\{\delta, \delta^2\}, \max\{\delta^2, \delta^4\}\right\}} &\leq 2 \exp\left(-\frac{n c'}{4K^4} \underbrace{\min\left\{\max\left\{\frac{\varepsilon^2}{n}, \frac{\varepsilon^4}{n^2}\right\}, \max\left\{\frac{\varepsilon}{\sqrt{n}}, \frac{\varepsilon^2}{n}\right\}\right\}}_{=\varepsilon^2/n}\right) \\ &= \delta^2 \end{aligned}$$

Exercise: Show that

$$\min\left\{\max\left\{\frac{\varepsilon}{\sqrt{n}}, \frac{\varepsilon^2}{n}\right\}, \max\left\{\frac{\varepsilon^2}{n}, \frac{\varepsilon^4}{n^2}\right\}\right\} = \frac{\varepsilon^2}{n} \quad \forall \varepsilon > 0.$$

Using the above exercise,

$$\mathbb{P}(|\|X\|_2 - \sqrt{n}| > \varepsilon) \leq 2 \exp\left(-\frac{c \varepsilon^2}{K^4}\right), \quad c = \frac{c'}{4}.$$

$$\Rightarrow \|X\|_2 - \sqrt{n} \text{ is sub-Gaussian}$$

$$\Rightarrow \exists \text{ universal } c' > 0 \text{ s.t.}$$

$$\|\|X\|_2 - \sqrt{n}\|_{\psi_2} \leq c' K^2.$$

Application: Dimensionality Reduction & Johnson-Lindenstrauss Lemma

- Consider vectors $\underline{a}_1, \dots, \underline{a}_N \in \mathbb{R}^D$, where $D \gg N$.
- Suppose that we wish to cluster $\underline{a}_1, \dots, \underline{a}_N$. Typically, clustering involves computing pairwise distances between the vectors.
- Complexity of computing pairwise distances = $O(N^2 D)$

- The idea of dimensionality reduction is to project $\underline{a}_1, \dots, \underline{a}_N$ to a lower dimensional space, say $\mathbb{R}^d$, where $d \ll D$, so that clustering may be performed in $\mathbb{R}^d$
- However, it is important to preserve the clustering after projection. That is, pairwise distances between the vectors should not alter much after projection.

Definition (ϵ -isometry)

Fix $\epsilon > 0$. A function $g: \mathbb{R}^D \rightarrow \mathbb{R}^d$ is called an ϵ -isometry if

$$(1-\epsilon) \|\underline{a}-\underline{b}\| \leq \|g(\underline{a})-g(\underline{b})\| \leq (1+\epsilon) \|\underline{a}-\underline{b}\|$$

Lemma (Johnson-Lindenstrauss)

Fix $\epsilon, \delta > 0$. If $g(\underline{a}_i) = \underline{X} \underline{a}_i \quad \forall i \in \{1, \dots, N\}$, where $\underline{X}$ is a $d \times D$ matrix with IID sub-Gaussian entries with common mean = 0, variance $\sigma^2 = \frac{1}{d}$,

and if $d = \Omega\left(\frac{K^4}{\epsilon^2} \log \frac{N}{\sqrt{\delta}}\right)$, then

The constant K will be shown in the proof

$$\mathbb{P}\left(\bigcup_{i \neq j} \left| \|\underline{X}(\underline{a}_i - \underline{a}_j)\| - \|\underline{a}_i - \underline{a}_j\| \right| > \epsilon \|\underline{a}_i - \underline{a}_j\| \right) \leq \delta.$$

Proof of the Johnson-Lindenstrauss lemma

Let $\underline{U}_i = \underline{X} \underline{a}_i$, $i \in \{1, \dots, N\}$. ; $\underline{a}_i = \begin{bmatrix} a_{i1} \\ \vdots \\ a_{iD} \end{bmatrix}$

$d \times D$ $\swarrow$

$$\underline{U}_i = \begin{bmatrix} \sum_{j=1}^D x_{1j} a_{ij} \\ \sum_{j=1}^D x_{2j} a_{ij} \\ \vdots \\ \sum_{j=1}^D x_{dj} a_{ij} \end{bmatrix}$$

$d \times 1$

$$\underline{U}_i - \underline{U}_j = \underline{X} (\underline{a}_i - \underline{a}_j) = \begin{bmatrix} \sum_{k=1}^D x_{1k} (a_{ik} - a_{jk}) \\ \vdots \\ \sum_{k=1}^D x_{dk} (a_{ik} - a_{jk}) \end{bmatrix}_{d \times 1}$$

Then,

$$E \left[\left\| \underline{U}_i - \underline{U}_j \right\|_2^2 \right] = \sum_{l=1}^d E \left[\left(\underbrace{\sum_{k=1}^D x_{lk} (a_{ik} - a_{jk})}_{\text{zero mean RV, iid across } k} \right)^2 \right]$$

$$= \sum_{l=1}^d \text{Var} \left(\sum_{k=1}^D x_{lk} (a_{ik} - a_{jk}) \right)$$

$$= \sum_{l=1}^d \sum_{k=1}^D (a_{ik} - a_{jk})^2 \underbrace{\text{Var}(x_{lk})}_{\text{iid with variance } \sigma^2}$$

$$= \sigma^2 \sum_{l=1}^d \sum_{k=1}^D (a_{ik} - a_{jk})^2$$

$$= \sigma^2 \sum_{l=1}^d \left\| \underline{a}_i - \underline{a}_j \right\|_2^2$$

$$= d \sigma^2 \left\| \underline{a}_i - \underline{a}_j \right\|_2^2$$

$$= \left\| \underline{a}_i - \underline{a}_j \right\|_2^2 \quad \left(\sigma^2 = \frac{1}{d} \right)$$

Write

$$\underline{X} = \frac{1}{\sqrt{d}} \underline{Y}, \quad \text{where entries of } \underline{Y} \text{ are IID, mean 0, Variance 1.}$$

Then, for fixed i, j such that $i \neq j$,

$$\mathbb{P} \left(\left| \left\| \underline{X} \underline{a}_i - \underline{X} \underline{a}_j \right\|_2 - \left\| \underline{a}_i - \underline{a}_j \right\|_2 \right| > \varepsilon \left\| \underline{a}_i - \underline{a}_j \right\|_2 \right)$$

$$= \mathbb{P} \left(\left| \left\| \frac{1}{\sqrt{d}} \underline{Y} \underline{a}_i - \frac{1}{\sqrt{d}} \underline{Y} \underline{a}_j \right\|_2 - \left\| \underline{a}_i - \underline{a}_j \right\|_2 \right| > \varepsilon \left\| \underline{a}_i - \underline{a}_j \right\|_2 \right)$$

$$= \mathbb{P} \left(\left| \left\| \frac{1}{\|\underline{a}_i - \underline{a}_j\|} \underline{y}(\underline{a}_i - \underline{a}_j) \right\|_2 - \sqrt{d} \right| > \varepsilon \sqrt{d} \right).$$

$\underline{y} = \underline{Z}$

Observe that $\underline{Z} = (Z_1, \dots, Z_d)$, where $Z_1, \dots, Z_d$ are iid,

$$\mathbb{E}[Z_1] = 0, \quad \text{Var}(Z_1) = 1.$$

$$\leq 2 \exp \left(-c \frac{(\varepsilon \sqrt{d})^2}{K^4} \right), \quad K = \|\underline{Z}_1\|_{\psi_2}$$

$$= 2 \exp \left(-\frac{c \varepsilon^2 d}{K^4} \right).$$

Therefore, by the union bound,

$$\mathbb{P} \left(\bigcup_{i \neq j} \left| \left\| \underline{x}(\underline{a}_i - \underline{a}_j) \right\| - \|\underline{a}_i - \underline{a}_j\| \right| > \varepsilon \|\underline{a}_i - \underline{a}_j\| \right)$$

$$\leq 2 \cdot N C_2 \cdot \exp \left(-\frac{c \varepsilon^2 d}{K^4} \right).$$

$$\leq N^2 \exp \left(-\frac{c \varepsilon^2 d}{K^4} \right).$$

Solving

$$N^2 \exp \left(-\frac{c \varepsilon^2 d}{K^4} \right) \leq \delta,$$

we get

$$d \geq \frac{K^4}{c \varepsilon^2} \log \frac{N^2}{\delta}$$

$$= \Omega \left(\frac{K^4}{\varepsilon^2} \log \frac{N}{\sqrt{\delta}} \right).$$

Other Concentration Inequalities

- We have seen concentration inequalities for sums of the form

$$\sum_{k=1}^n X_k, \quad X_1, \dots, X_n \text{ independent.}$$

- We will now consider

$$Z_n = g(X_1, \dots, X_n), \quad X_1, \dots, X_n \text{ independent}$$

for some general $g: \mathbb{R}^n \rightarrow \mathbb{R}$.

- If $X_1, \dots, X_n$ are independent, then

$$\text{Var}\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n \text{Var}(X_k).$$

Can we derive a similar relation for $\text{Var}(g(X_1, \dots, X_n))$?

Given $i \in \{1, \dots, n\}$, let

$$\Delta_i = \underbrace{\mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_i]}_{Z_i} - \underbrace{\mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_{i-1}]}_{Z_{i-1}}.$$

By convention,

$$Z_0 = \mathbb{E}[g(X_1, \dots, X_n)], \quad Z_n = g(X_1, \dots, X_n).$$

Then,

$$\begin{aligned} \sum_{i=1}^n \Delta_i &= Z_1 - Z_0 + Z_2 - Z_1 + \dots + Z_n - Z_{n-1} \\ &= Z_n - Z_0 \\ &= g(X_1, \dots, X_n) - \mathbb{E}[g(X_1, \dots, X_n)]. \end{aligned}$$

We then have

$$\begin{aligned} \text{Var}(g(X_1, \dots, X_n)) &= \mathbb{E}\left[\left(\sum_{i=1}^n \Delta_i\right)^2\right] \\ &= \mathbb{E}\left[\sum_{i=1}^n \Delta_i^2\right] + \mathbb{E}\left[\sum_{i \neq j} \Delta_i \Delta_j\right] \end{aligned}$$

$$= \mathbb{E} \left[\sum_{i=1}^n \Delta_i^2 \right] + 2 \underbrace{\mathbb{E} \left[\sum_{i < j} \Delta_i \Delta_j \right]}.$$

For $i < j$:

$$\begin{aligned} \mathbb{E}[\Delta_i \Delta_j] &= \mathbb{E} \left[\mathbb{E}[\Delta_i \Delta_j \mid X_1, \dots, X_i] \right] \\ &= \mathbb{E} \left[\Delta_i \mathbb{E}[\Delta_j \mid X_1, \dots, X_i] \right] \end{aligned}$$

Recall :

$$\Delta_j = Z_j - Z_{j-1}$$

$$= \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_j] - \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_{j-1}]$$

$$\mathbb{E}[\Delta_j \mid X_1, \dots, X_i]$$

$$\begin{aligned} &= \mathbb{E} \left[\mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_j] \mid X_1, \dots, X_i \right] \\ &\quad - \mathbb{E} \left[\mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_{j-1}] \mid X_1, \dots, X_i \right] \end{aligned}$$

$$\mathbb{E}[\mathbb{E}[Z \mid X, Y] \mid Y] = \mathbb{E}[Z \mid Y] \quad (\text{Tower property})$$

$$= \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_i] - \mathbb{E}[g(X_1, \dots, X_n) \mid X_1, \dots, X_i]$$

$$= 0.$$

Therefore,

$$\begin{aligned} \text{Var}(g(X_1, \dots, X_n)) &= \sum_{i=1}^n \mathbb{E}[\Delta_i^2] \\ &= \sum_{i=1}^n \text{Var}(\Delta_i). \end{aligned}$$